CLINISCAN: LUNG-ABNORMALITY DETECTION ON CHEST X-RAYS USING AI

-INFOSYS SPRINGBOARD

Milestone 2: Data Preparation & Environment Setup

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1.INTRODUCTION

The objective of Milestone 2 is to develop and implement the core AI components for detecting and classifying lung abnormalities from chest X-ray images. This phase transitions from data preparation to active model development, establishing the baseline performance for the CliniScan system.

In this milestone, we implemented two primary models:

• Detection Model: Designed to identify and locate specific abnormalities using bounding boxes.

• Classification Model: Designed to predict the overarching disease category from the X-ray images

2. DATA DESCRIPTION

The system utilizes the VinDr-CXR dataset, which consists of 15,000 annotated chest X ray images for training.

Key Data Characteristics:

Annotations: Provided in train.csv, containing disease labels and bounding box coordinates.
Metadata: Image dimensions (height and width) are stored in train_meta.csv.
Image Format: Images were processed in resized PNG format at a resolution of 1024×1024.

3. DATA PREPROCESSING

To prepare the data for training, several cleaning and transformation steps were performed:
Filtering: Removed "No Finding" entries and rows with missing (NaN) bounding boxes to focus the detection model on valid abnormalities.

YOLO Conversion: Bounding boxes were normalized into YOLO format using the equations
Empty Labels: Generated empty label files for images without abnormalities to maintain dataset integrity.

Verified dataset integrity

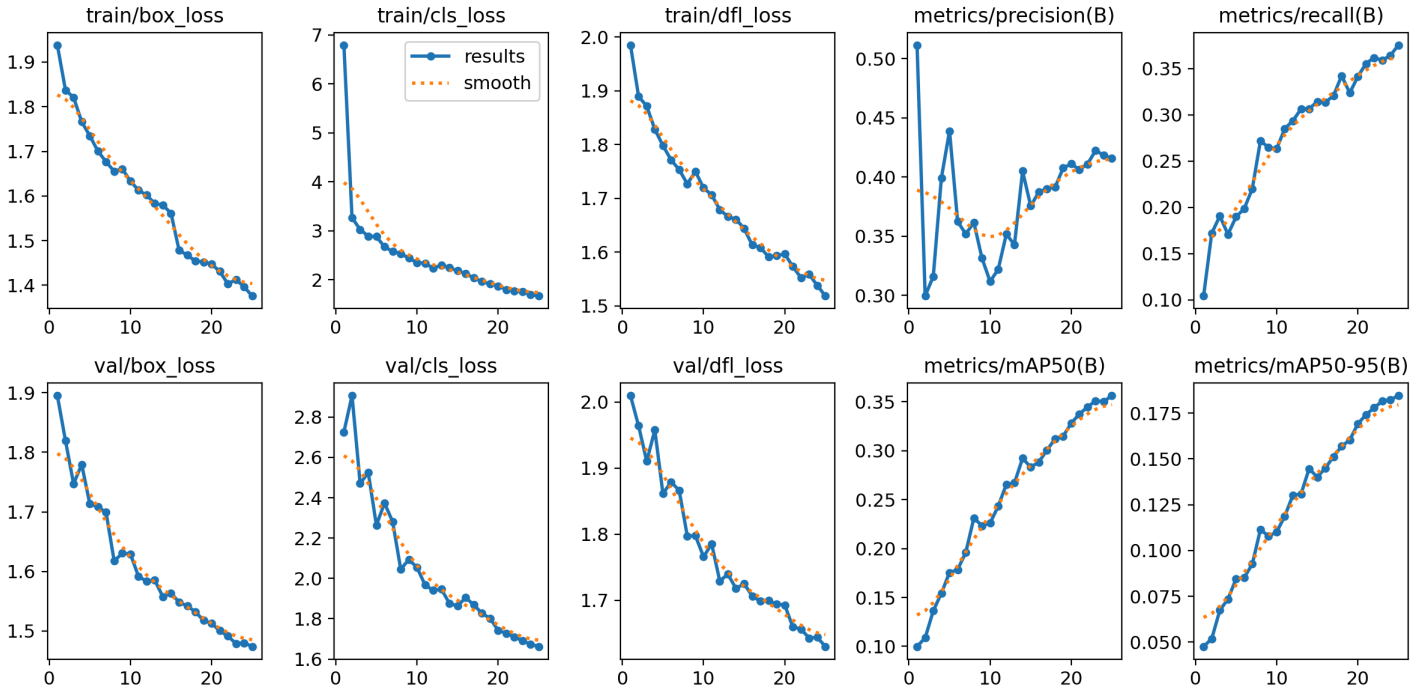
Total images:15000

Total labels:15000

4. DETECTION MODEL(YOLOv11)

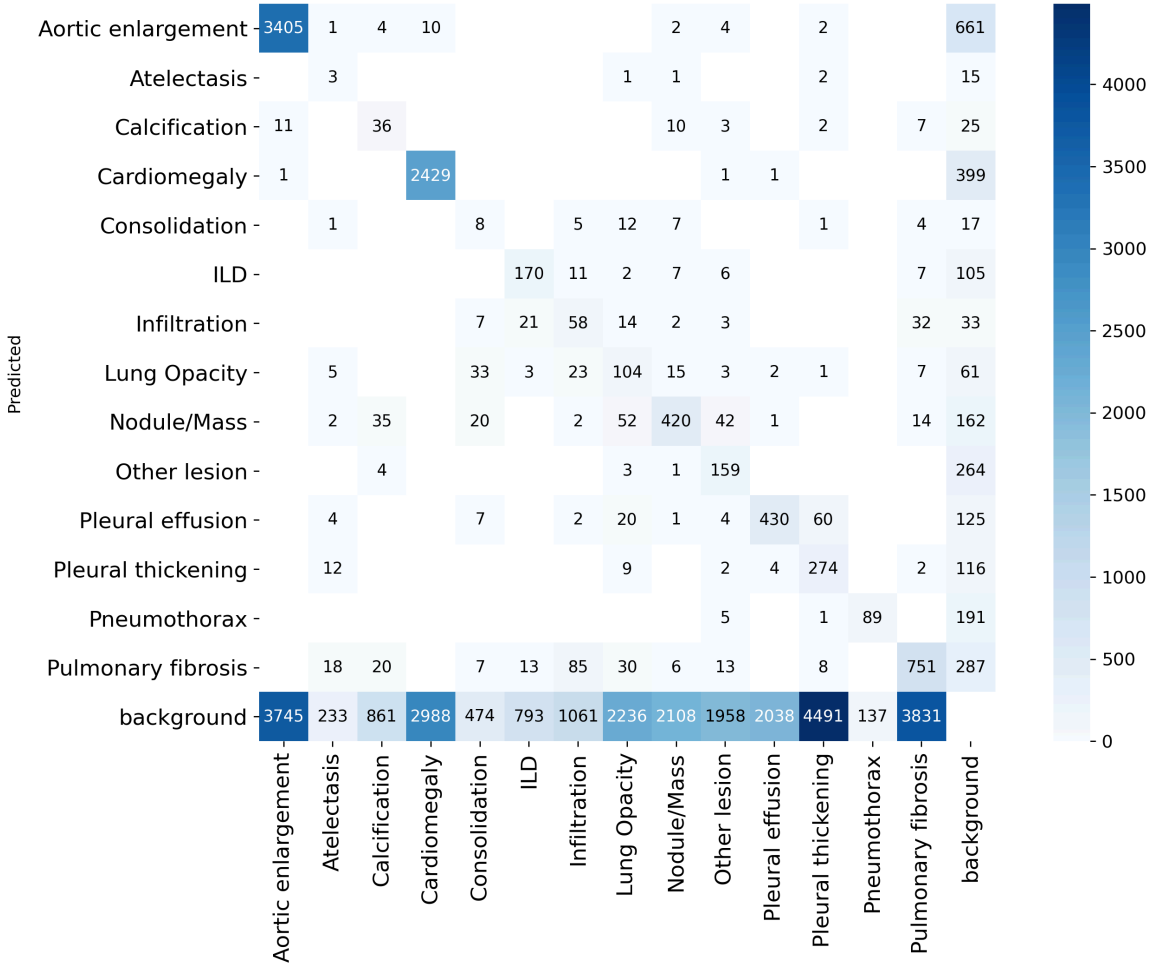
We utilized the YOLOv11 architecture from Ultralytics for the detection task.
Training Configuration:
Input Size: 1204*1204
Echos:25
Batch Size: 8
Classes: 14 disease categories

Training Visualizations:

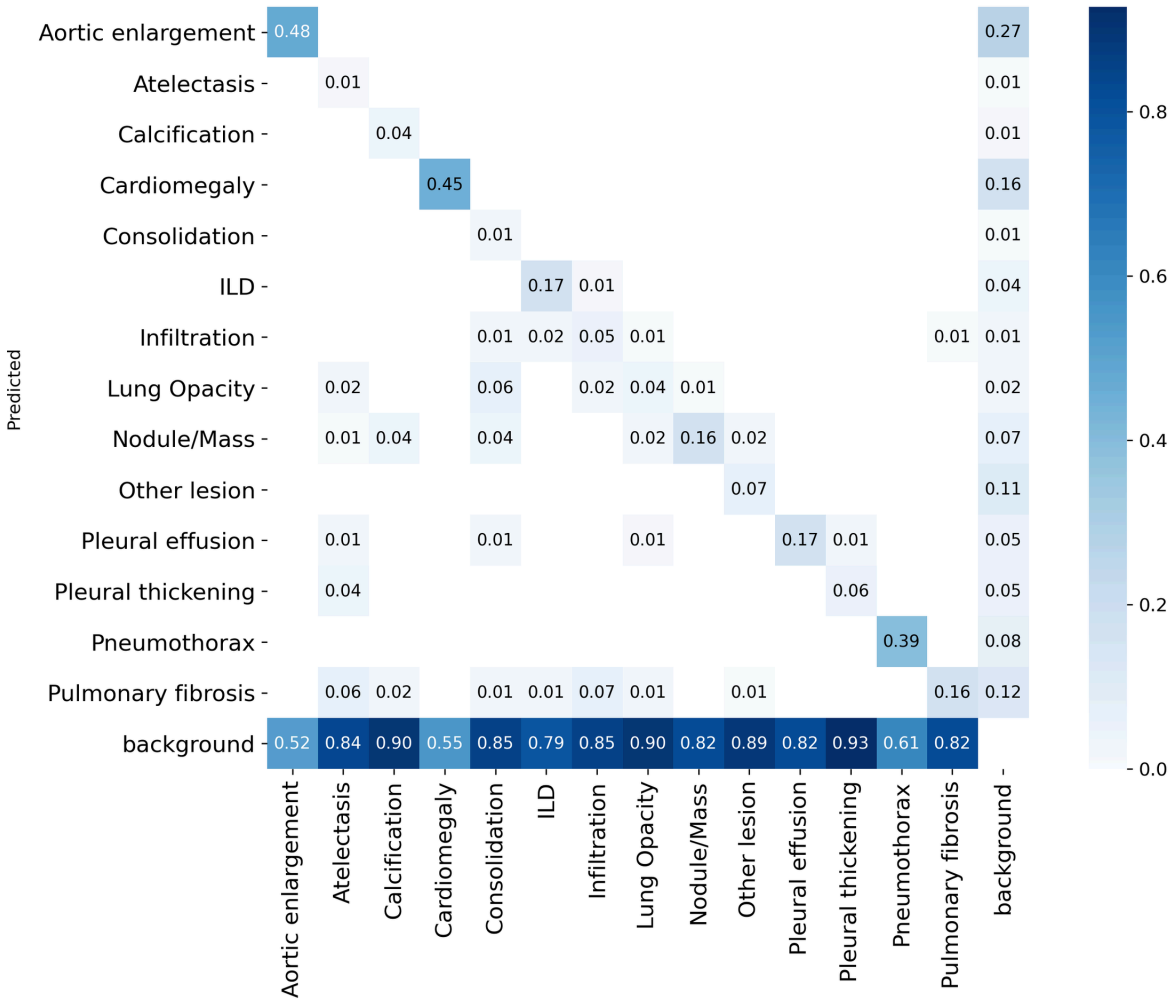


5. CLASSIFICATION MODEL (ResNet50)

For the classification task, a ResNet50 model pretrained on ImageNet was implemented.
Approach: To establish a baseline, the multi-label dataset was converted into a single label classification task by selecting one primary disease per image



Confusion Matrix Normalized



6. SYSTEM WORKFLOW & MODEL SAVING

The integrated pipeline for CliniScan is structured as follows:

Input: Chest X-ray Image
Detection: YOLO model locates the abnormal region
Classification: ResNet model predicts the specific disease type
Saved Artifacts:
• Detection Model: cliniscan_detection_model.pt
• Classification Model: cliniscan_classification_model.pth

Challenges Faced:

• Handling multi-label data and class imbalances.
• GPU limitations within the colab environment.
• Incorrect bounding box normalization

7.REFERENCES

• Vin-CXR Dataset
• Ultralytics YOLO Documentation
• Py Torch Documentation